Syllabus (Lecture)

1. Static Routing
2. Dynamic Routing
3. Virtual LAN (VLAN)
4. Spanning Tree Protocol (STP)
5. Ethernet Link Aggregation
6. Access Control List (ACL)
7. Authentication, Authorization және Accounting (AAA)
8. Network Address Translation (NAT)
9. Network Services (FTP, TFTP, Telnet, DHCP, HTTP, DNS, NTP)
10. Wireless LAN (WLAN)
11. WAN хаттамалары
12. IPv6 addressing
13. Network Automation. Python
14. Network Automation. Ansible
15. Қорытынды

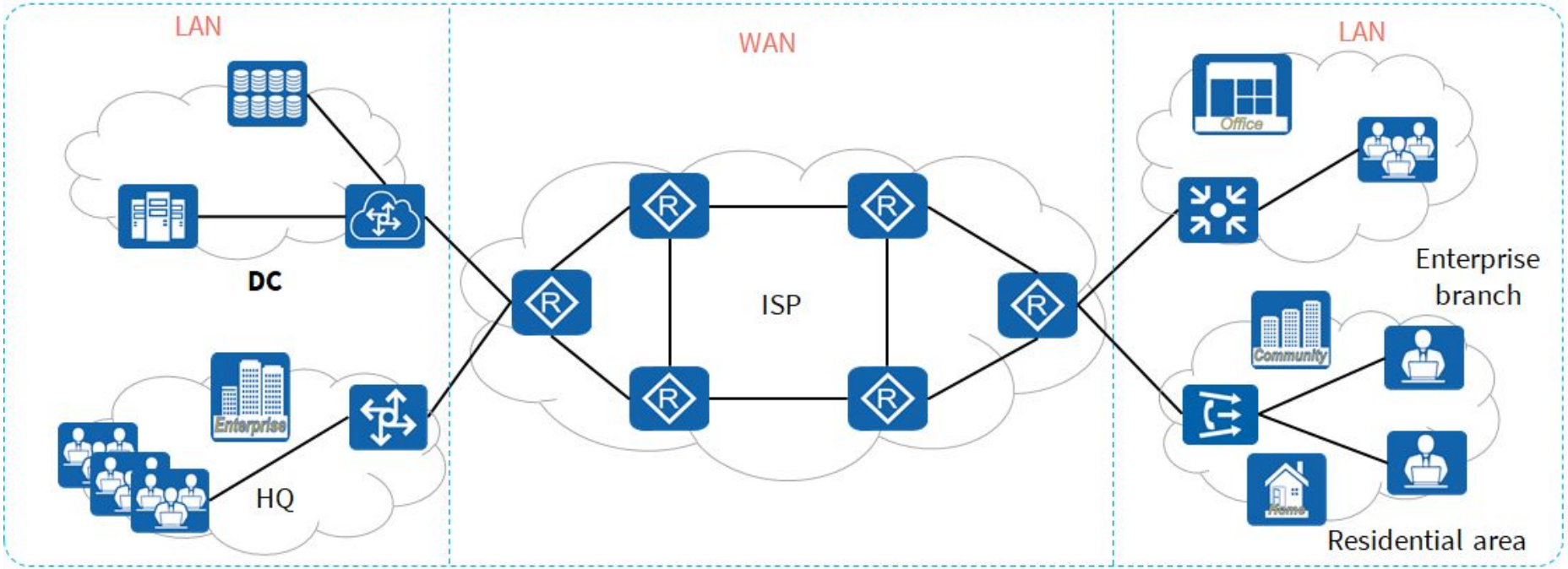
Lesson #11

by Kaiyrkhan Mukapil

- ➔ WAN хаттамалары
  - ✔ HDLC
  - ✔ PPP
    - ☑️ PPP PAP Authentication
    - ☑️ PPP CHAP Authentication
  - ✔ PPPoE
  - ✔ Frame Relay

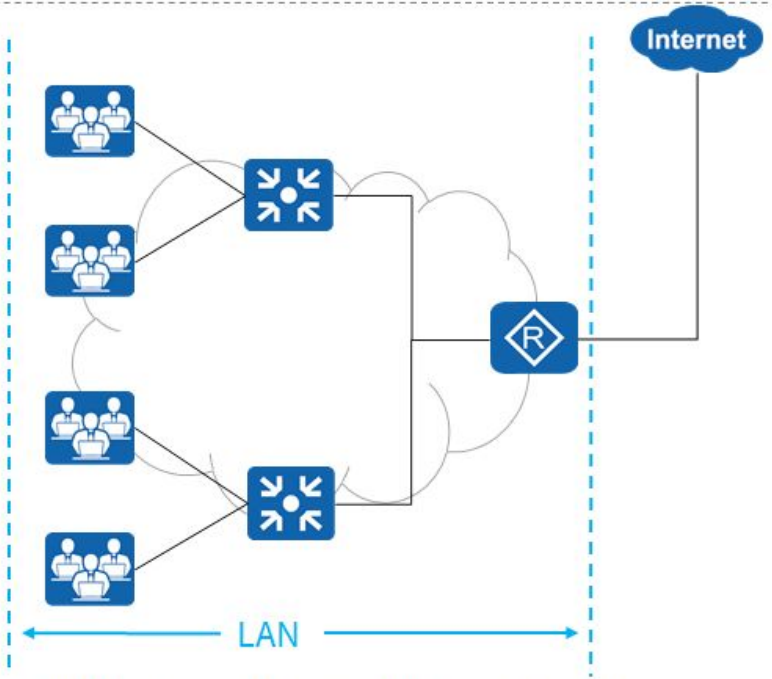
What Is a WAN?

A WAN is a network that connects LANs in different areas. A WAN generally covers tens of kilometers to thousands of kilometers. It can connect multiple regions, cities, and countries, or provide long-distance communication across several continents, forming an international remote network.



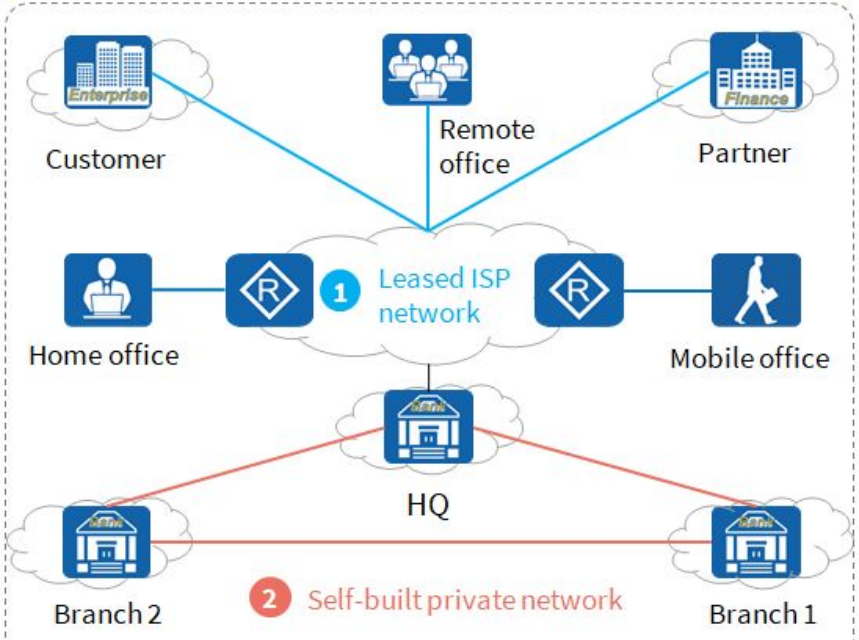
Differences Between a WAN and a LAN

LAN



- A LAN is a computer network that covers a small geographical area.

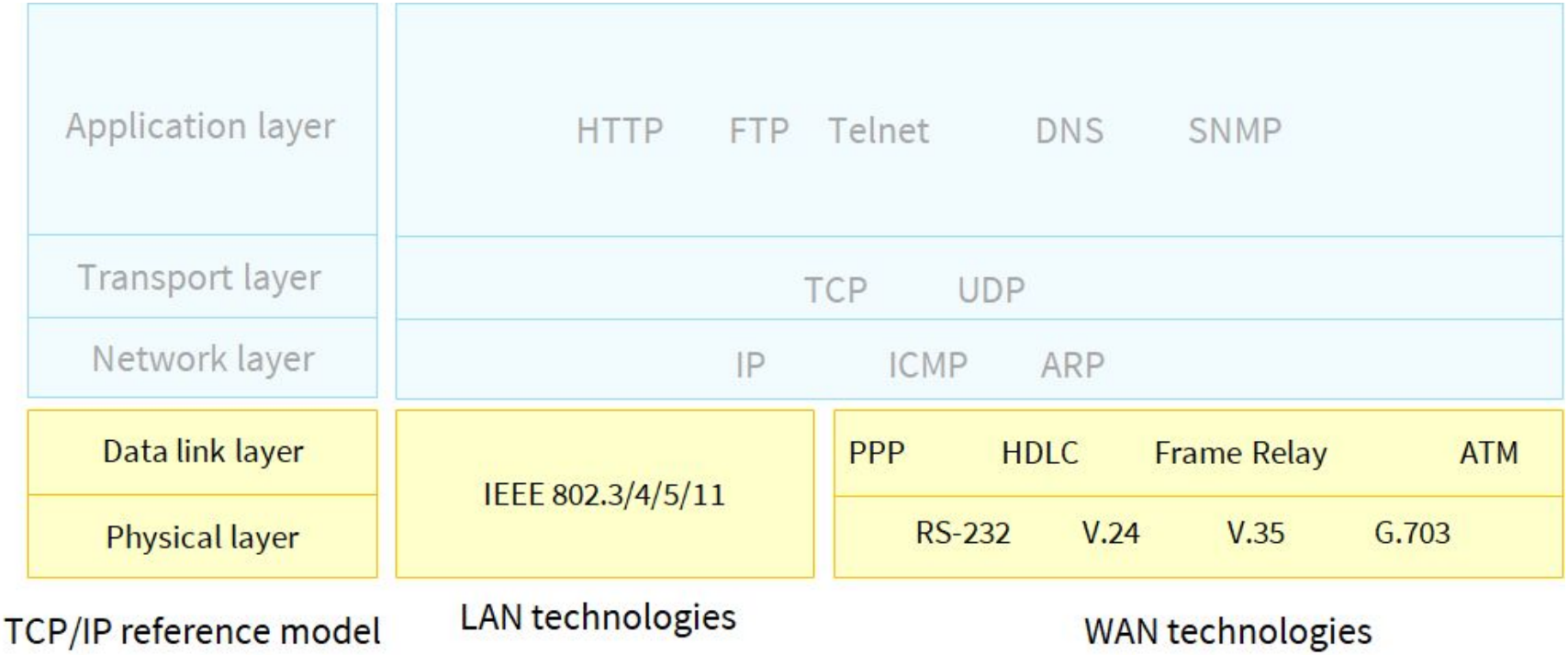
WAN



- A WAN is a computer network that covers a wide area by leasing an Internet service provider (ISP) network or building a private network.

Overview of Early WAN Technologies

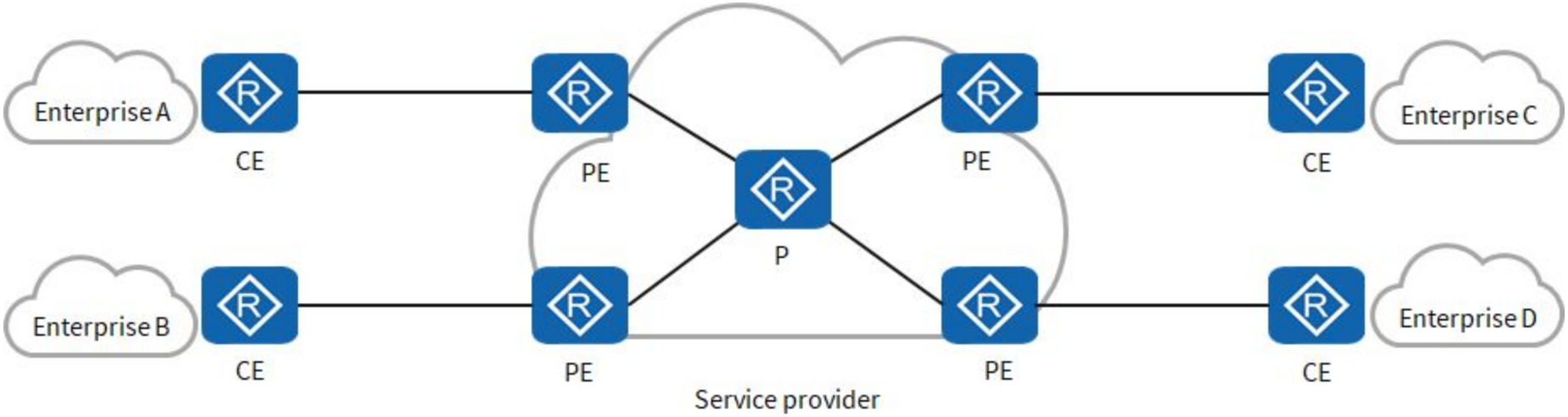
The early WANs and LANs differ in the data link layer and physical layer and are the same in the other layers in the TCP/IP reference model.



WAN Device Roles

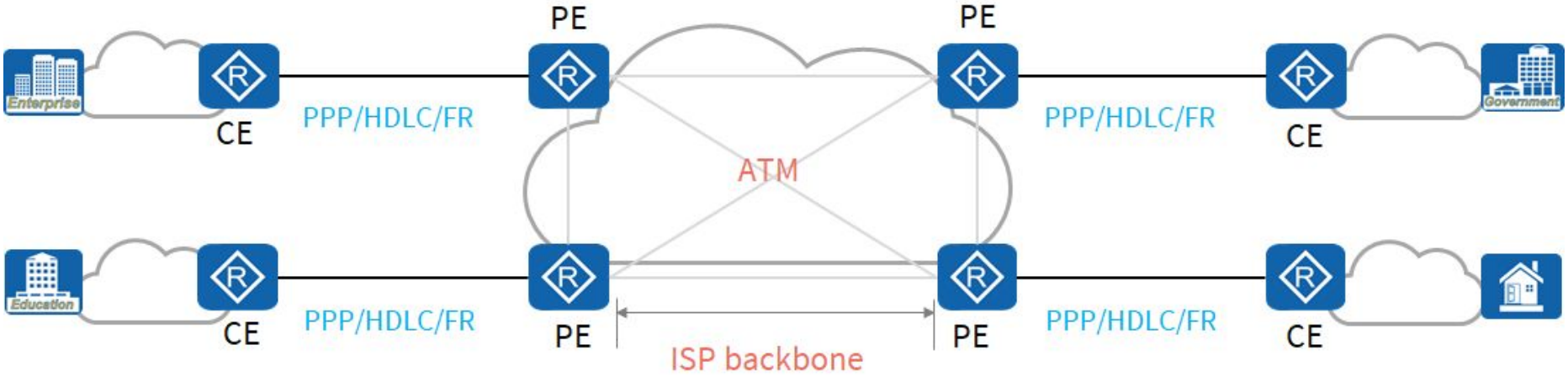
There are three basic roles of WAN devices: customer edge (CE), provider edge (PE), and provider (P). They are defined as follows:

- ✅ CE: a device located at the customer premises and connected to one or more PEs for user access.
- ✅ PE: a service provider's important edge device that is connected to both a CE and a P.
- ✅ P: a service provider's device that is not connected to any CE.



Application of Early WAN Technologies

The early WAN technologies perform different Layer 2 encapsulation at the data link layer for different types of physical links. PPP, HDLC, and FR are commonly used between CEs and PEs to implement long-distance transmission of user access packets over a WAN. ATM is commonly used on ISP backbone networks for high-speed forwarding.



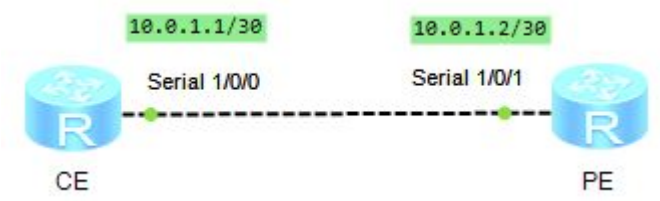
Configuring HDLC encapsulation

CE, PE

```
display interface Serial 1/0/0
Link layer protocol is PPP

interface Serial 1/0/0
link-protocol hdlc
ip address 10.0.1.1 30
display this

display interface Serial 1/0/0
Link layer protocol is nonstandard HDLC

display ip int brief
```

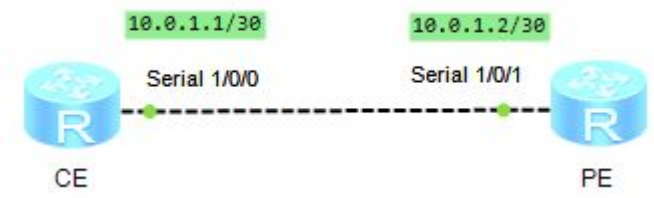
Ping from R1 to R2

```
ping 10.0.1.2
```



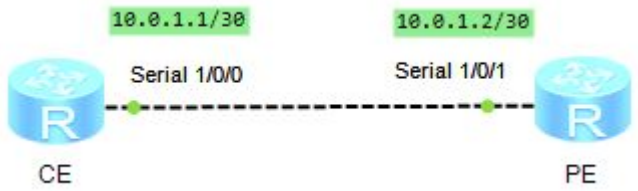
Configuring PPP encapsulation with PAP Authentication

```
# CE

interface Serial 1/0/0
link-protocol ppp
ip address 10.0.1.1 30

interface Serial 1/0/0
ppp authentication-mode pap

aaa
local-user user1 password cipher user@123
local-user user1 service-type ppp
display this
```



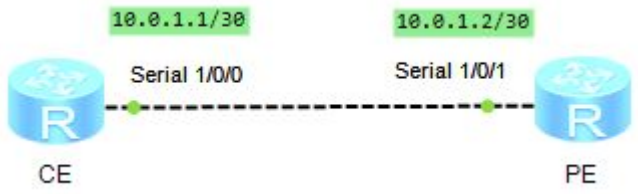
Configuring PPP encapsulation with PAP Authentication

```
# PE

interface Serial 1/0/1
link-protocol ppp
ip address 10.0.1.2 30

interface Serial 1/0/1
ppp pap local-user user1 password cipher user@123
```



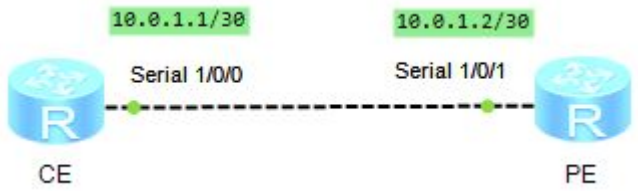
Configuring PPP encapsulation with CHAP Authentication

```
# CE

interface Serial 1/0/0
link-protocol ppp
ip address 10.0.1.1 30

interface Serial 1/0/0
ppp authentication-mode chap

aaa
local-user user1 password cipher user@123
local-user user1 service-type ppp
display this
```



Configuring PPP encapsulation with CHAP Authentication

```
# PE

interface Serial 1/0/1
link-protocol ppp
ip address 10.0.1.2 30

interface Serial 1/0/1
ppp chap user user1
ppp chap password cipher user@123
display this
```



Configuring PPPoE

PPPoE Server

Create a pool address

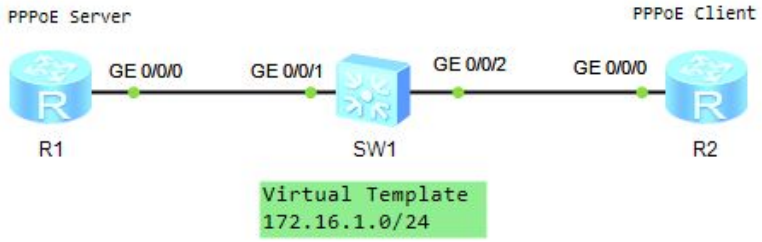
```
ip pool POOL1
network 172.16.1.0 mask 24
gateway-list 172.16.1.1
```

Create a Virtual Template interface

```
interface Virtual-Template 1
ip address 172.16.1.1 24
ppp authentication-mode chap
remote address pool POOL1

display ip pool
```

Bind the Virtual Template

```
int g0/0/0
pppoe-server bind virtual-template 1
```

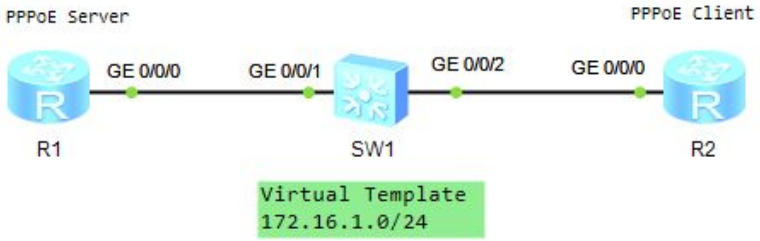


Configuring PPPoE

PPPoE Server

Create a Local User

```
aaa
local-user user1 password cipher user@123
local-user user1 service-type ppp
```



Configuring PPPoE

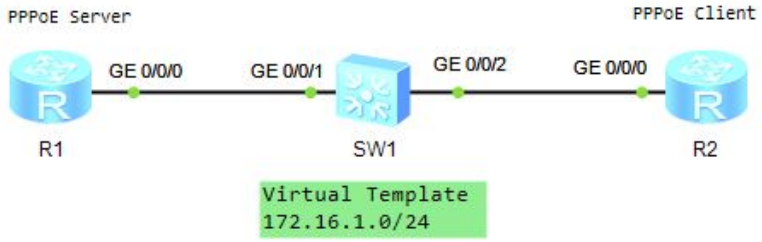
PPPoE Client

Create a dialer-rule

```
dialer-rule
dialer-rule 1 ip permit

interface Dialer 1
dialer user u1
dialer-group 1
dialer bundle 1
ppp chap user user1
ppp chap password cipher user@123
ip address ppp-negotiate

int g0/0/0
pppoe-client dial-bundle-number 1
```

Create a default static route

```
ip route-static 0.0.0.0 0.0.0.0 Dialer 1
```

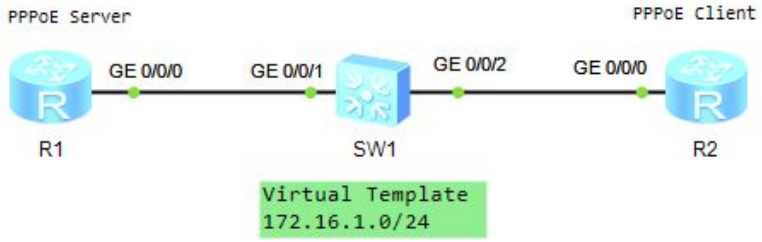
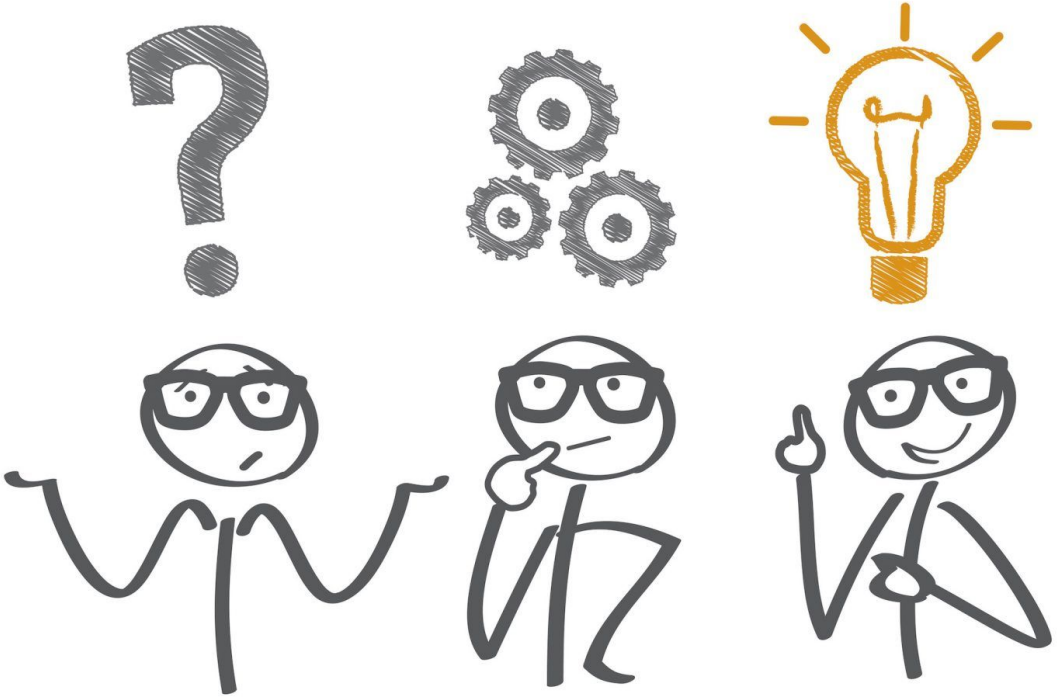


Configuring PPPoE

PPPoE Server

```
display pppoe-server session all
```

PPPoE Client

```
display ip int brief
ping 172.16.1.1
```

```
display interface Dialer 1
display pppoe-client session summary
```